ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Main)
TEST # 04
18-08-2024

JEE(Main+Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Review (Unit Test # 01, 02 & 03)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	B	C	C	D	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	A	B	D	D	C	A	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8	8	1	5	2	0	4	2	2	9

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	A	A	B	A	A	A	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	D	D	C	C	B	C	D	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9	7	1	314	2	3	4	7	4	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	C	D	C	A	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	C	C	A	C	B	D	B	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	26	6	4	1	8	2	0	2	9	6

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

$\sin 480 = \sin(360 + 120)$

$= \sin 120 = \sin(90 + 30)$

$= \cos 30 = \frac{\sqrt{3}}{2}$

2. Ans (B)

$\left[\frac{ax}{t}\right] = [M^0L^0T^0]$

$[a]\left[\frac{L}{T}\right] = [M^0L^0T^0]$

$[a] = [L^{-1}T]$

4. Ans (B)

$T = \sqrt{\frac{2h}{10}} = 2\,\text{sec}$

$a_x = \frac{f}{m} = \frac{8}{2} = 4\text{m/s}$

$x = \frac{1}{2}a \times T^2 = \frac{1}{2} \times 4 \times 2^2 = 8\text{m}$

8. Ans (B)

$(V_y)_1 = (V_y)_2$

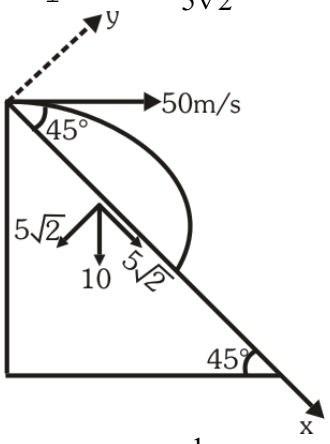
$V_1 \sin 30° = V_2$

$\frac{V_1}{V_2} = \frac{1}{\sin 30°} = 2$

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9. Ans (B)

$$T = \frac{2u_\perp}{a_\perp} = \frac{2 \times 25\sqrt{2}}{5\sqrt{2}} = 10s$$



$$R = (25\sqrt{2})(10) + \frac{1}{2}(5\sqrt{2})(10)^2 = 500\sqrt{2}m$$

13. Ans (A)

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, u = −60, f = −30

$\frac{1}{-60} + \frac{1}{v} = -\frac{1}{30}, \frac{1}{v} = -\frac{1}{30} + \frac{1}{60} = -\frac{1}{60}$

v = −60 cm and , $\frac{1}{v^2}\frac{dv}{dt} + \frac{1}{u^2}\frac{du}{dt} = 0$

$\frac{du}{dt} = -\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$

$\frac{dv}{dt}$ =−5m/s

14. Ans (B)

The position of image of nearer end to mirror can be found out using

mirror equation, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{v_1} = \frac{1}{-20} - \frac{1}{-10}$

$\Rightarrow v_1 = -20$

Similarly position of image of farther end to mirror can be found out as

$= \frac{1}{v_2} = \frac{1}{-(10+20)} - \frac{1}{-10}$

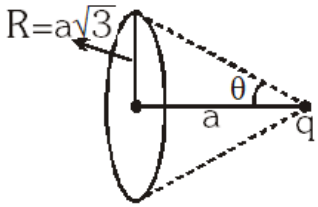
$\Rightarrow v_2 = -15$

Hence the length of the image

= 20cm − 15cm = 5cm

16. Ans (D)

We know that if a point charge q is placed on the apex of a cone of semi-vertex angle θ then



electric flux through the base of the cone will be

$\frac{q(1-\cos\theta)}{2\epsilon_0}$ here $\tan\theta = \frac{a\sqrt{3}}{a} = \sqrt{3} \Rightarrow \theta = 60°$

Therefore $\phi = \frac{q(1-\cos 60°)}{2\epsilon_0} = \frac{q}{4\epsilon_0} \Rightarrow q = 4\epsilon_0\phi$

17. Ans (C)

If there is a cavity inside the conductor having no charge then charge will always reside only on outer surface of conductor.

18. Ans (A)

As, $v_e = \sqrt{\left(\frac{2GM}{R}\right)}, R = \left(\frac{2GM}{v_e^2}\right)$,

$\therefore R = \frac{2 \times 6.67 \times 10^{-11} \times 6 \times 10^{24}}{(3 \times 10^8)^2} = 9 \times 10^{-3}m = 9mm$

19. Ans (B)

$T^2 \propto R^3$

$\left(\frac{T'}{T}\right)^2 = \left(\frac{9R}{R}\right)^3$

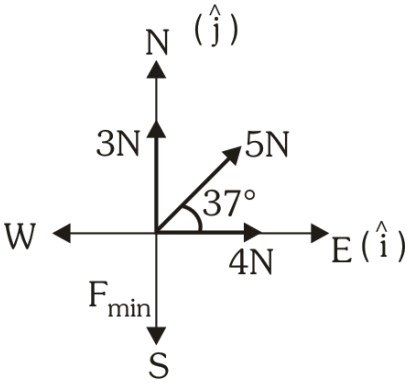
$T'^2 = T^2 \times 9^3$

$T' = T \times 3^3$

$T' = 27\ T$

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20. Ans (B)



Let force be F so resultant is in east direction

$4\hat{i} + 3\hat{j} + \left(5\cos 37^\circ \hat{i} + 5\sin 37^\circ \hat{j}\right) + \vec{F} = k\hat{i}$

$\Rightarrow 4\hat{i} + 3\hat{j} + 4\hat{i} + 3\hat{j} + \vec{F} = k\hat{i}$

$\Rightarrow 8\hat{i} + 6\hat{j} + \vec{F} = k\hat{i}$

$\Rightarrow \vec{F} = (k-8)\hat{i} - 6\hat{j}$

$\Rightarrow F = \sqrt{(k-8)^2 + (6)^2} \Rightarrow F_{min} = 6N$

PART-1 : PHYSICS

SECTION-II

1. Ans (8)

$\vec{S}$ = displacement

$= (1-2)\hat{i} + (4-3)\hat{j} = -\hat{i} + \hat{j}$

W_1 = work done by force

$F_1 = \vec{F}_1 \cdot \vec{S} = \left(2\hat{i} + 3\hat{j}\right) \cdot \left(-\hat{i} + \hat{j}\right)$

$= -2 + 3 = 1$ joule.

W_2 = work done by force

$F_2 = \vec{F}_2 \cdot \vec{S} = \left(-3\hat{i} + 4\hat{j}\right) \cdot \left(-\hat{i} + \hat{j}\right)$

$= 3 + 4 = 7$ joule

$W_{net} = W_1 + W_0$

$= 7 + 1 = 8$ joule.

2. Ans (8)

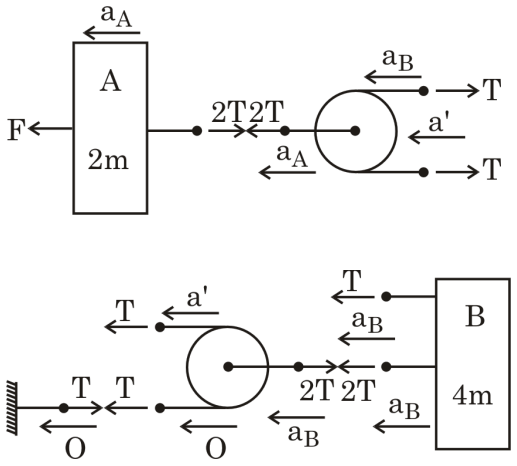
$a = \dfrac{20t - 80}{10} = 2t - 8$

$f_s = 5\,(2t-8) = f_L = 40$

$2t - 8 = 8$

$t = 8s$

3. Ans (1)



$a_B + a' = 2a_A$ [constrained relation for pulley 1]

$O + a' = 2a_B$ [constrained relation for pulley 2]

From above two equations

$3a_B = 2a_A$

$F - 2T = 2ma_A$ [Netwon's II law for block A]

.....II

$3T = 4\ m\ a_B$ [Netwon's II law for block

B] III

From equation I, II and III

$a_B = \dfrac{3F}{17m}$

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4. Ans (5)

for C $\Rightarrow 2T = mg$

$T = \frac{mg}{2}$

If C doesn't move then $a_A = 4a_B$ … (i)

$P - T - \mu mg = m\,4a_B$ …(ii)

$4T - mg = ma_B$ …(iii)

$\therefore \frac{P}{mg} = 5$

5. Ans (2)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{\mu}{2R} - \frac{1}{\infty} = \frac{\mu - 1}{R}$$

$$\frac{\mu}{2R} = \frac{\mu - 1}{R}$$

$\mu = 2$

6. Ans (0)

$E_B = 0 \Rightarrow \frac{E_B}{E_A} = 0$

7. Ans (4)

For small drops $V = \frac{kq}{r} = 1$ volt

For bigger drop

$V' = \frac{k(8q)}{2r} = 4\left(\frac{kq}{r}\right) = 4$ volt

9. Ans (2)

P.E = 2 (T.E)

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

$M_{avg} = \frac{95 \times 12 + 5 \times 14}{100} = 12.1$

No. of proton = 6

So no. of neutrons = 6.1

2. Ans (D)

O_3 O_2

Mole ratio : 1 2

$[VD]_{mix} = \frac{M_{avg}}{2} = \frac{48 + 64}{2 \times 3} = \frac{112}{2 \times 3} = \frac{56}{3}$

3. Ans (A)

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(\ell)$

x ml (55−x)ml

LR

Volume contraction

$= V_{used} - V_{produced}$

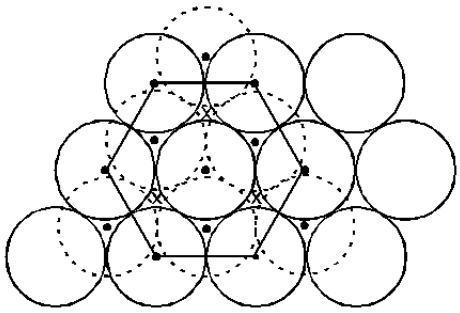
$45 = \left(x + \frac{x}{2}\right) - 0$

x = 30 ml

4. Ans (A)

Fact

5. Ans (B)



hcp = AB AB AB .....pattern repeat

For calculating voids between two layers A and B.

The dots are representing tetrahedral voids.

no. of tetrahdral voids $= 3 + 1 + \frac{1}{3} \times 6 = 6$

so in one complete unit cell total tetrahedral voids = 12, out of which 8 are completely inside but rest are shared by other unit cells.

Octahedral voids = 3 = represented by (x).

So total octahedral voids = 6 = All are completely inside.

[$4R = \sqrt{3}\,a$]

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6. **Ans (A)**

$$\frac{P^0 - P_s}{P_s} = \frac{n_{B\ solute}}{n_{A\ solvent}}$$

$$\frac{P^0 - 20}{20} = \frac{6}{M} \times \frac{18}{180} \quad \text{.....(1)}$$

$$\frac{P^0 - 20.02}{20.02} = \frac{6}{M} \times \frac{18}{198}$$

$P^0 = 20.22$ torr.

7. **Ans (A)**

$$CH_3CHO(g) \longrightarrow CH_4(g) + CO(g)$$

t = 0	80	0	0
t = 20min	80 – x	x	x

$(80 - x) + x + x = 120 \Rightarrow x = 40$

$t_{1/2} = 20$min

$$K = \frac{0.693}{20} = 3.465 \times 10^{-2}\ min^{-1}$$

8. **Ans (A)**

For adiabatic process $P \propto T^3$

$$\left(\frac{P_1}{P_2}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T_1}{T_2}$$

$(P)^{\frac{\gamma-1}{\gamma}} \propto T$

$P \propto T^{\left(\frac{\gamma}{\gamma-1}\right)}$

According the question pressure is proportional to the cube of its absolute temperature hence

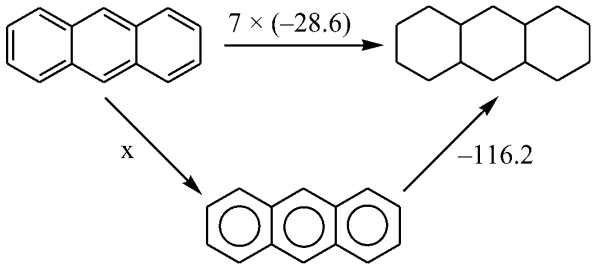
$$\frac{\gamma}{(\gamma - 1)} = 3$$

$\Rightarrow \gamma = 3\gamma - 3$

$2\gamma = 3$

$$\Rightarrow \gamma = \frac{C_P}{C_V} = \frac{3}{2}$$

10. **Ans (A)**



$-200.2 = x + (-116.2)$

$x = -200.2 + 116.2$

$$= -84\ \frac{kcal}{mol}$$

11. **Ans (C)**

Maximin electron in P-shell (n = 6)

$\Rightarrow 2n^2$

$\Rightarrow 2\times(6)^2 = 72$

12. **Ans (B)**

${}^{y}_{x}A$ & ${}^{q}_{p}B$ are isobar

$\therefore y = q$ ..... (1)

${}^{n}_{m}C$ & ${}^{y}_{x}A$ is are inotune

$\therefore (n - m) = (y - x)$ ..... (2)

From eq. (1) & (2)

$n - m = q - x$

$n = m + q - x$

13. **Ans (D)**

$$X^+ \xrightarrow{\text{IE of } X^{\oplus}} X^{+2}$$

$$X^+ \xleftarrow{|\Delta H_{eg}| \text{ of } X^{+2}} X^{+2}$$

$|\Delta H_{eg}|$ of X^{+2} = I.E. of X^+ = 4ev

14. **Ans (D)**

BONUS

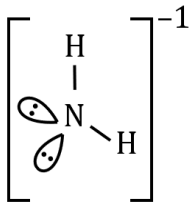
15. **Ans (C)**

Due to poor screening the I.E. increases

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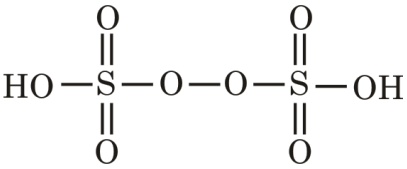
16. **Ans (C)**

NH_2^- ⇒ 2 b. p. +2 ℓ. p.



Planar & polar

19. **Ans (D)**



20. **Ans (D)**

no. of oxygen corrers shared per tetrahedral unit.

For 2D silicate ⇒ 3

For 3 D silicate ⇒ 4

For pyrosilicate = 1

PART-2 : CHEMISTRY

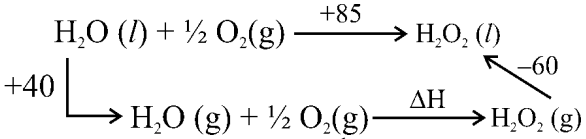
SECTION-II

1. **Ans (9)**

$H_2(g) + O_2(g) \longrightarrow H_2O_2(l)$; $\Delta H = -200$ .....(1)

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$; $\Delta H = -285$ .....(2)

Eq. (1) & Eq (2)



$85 = 40 + (2 \times \Delta H_{O-H} + \frac{1}{2} \times \Delta H_{O=O} - 2 \times \Delta H_{O-H} - \Delta H_{O-O}) - 60$

$\Delta H_{O-O} = 45$ kJ/mol

2. **Ans (7)**

q = 0 ; w = 0 ; $\Delta U = 0 = T \longrightarrow$ constant

$\Delta G = nRT \ln \frac{P_2}{P_1} = P_1V_1 \ln \frac{P_2}{P_1}$

$= 1 \times 10 \ln \frac{20}{10} = 10 \ln 2$

$= 10 \times 0.7 = 7$ bar.L

3. **Ans (1)**

* FeO crystallizes in NaCl structure $Fe^{2+} \longrightarrow$ octahedral voids

* $Fe_{0.75}O$ $O^{2-} \longrightarrow$ FCC lattice

* Assume 100 molecule crystal of $Fe_{0.75}O$

No. of Fe atom = 75

No. of O^{2-} atom = 100

By balancing the charge = 100 × 2 = 75 Fe

$Fe = \underset{x}{Fe^{2+}} , \underset{y}{Fe^{3+}}$

$x + y = 75$ .....(1)

$2x + 3y = 200$ .....(2)

$y = 50$

$x = 25$

* 100 molecule of $Fe_{0.75}O$ = 25 unit cells

* 4 O^{2-} occupy 1 unit cell

25 Fe^{2+} present in 25 unit cells = So 1 Fe^{2+} per unit cell

So Answer is (1)

4. **Ans (314)**

$\Delta T_f = 273 - 263 = 10$ °C

$\Delta T_f = i\, k_f .\, m$

$10 = \frac{1 \times 1.86 \times 62 \times 1000}{62 \times w_{solvent}}$

$w_{solvent} = 186$ g

Mass of ice freezed = 500 − 186 = 314 g

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5. **Ans (2)**

40 gm NaOH present in 100 gm of solution

Volume of solution = $\frac{100\text{ gm}}{2\text{ gm}} \times \text{ml} = 50\text{ ml}$

$M = \frac{40/40}{50/1000} = 20M$

$m = \frac{1}{(100-40)} \times 1000 = \frac{50}{3} m$

6. **Ans (3)**

$E \propto \frac{1}{r^3}$ for dipole – dipole interaction so pairs are (i), (iii) and (iv)

7. **Ans (4)**

K_2O_2 , $NO[BF_4]$, KCN, Na_2C_2
(All are diamagnetic)

8. **Ans (7)**

$Zn_x(OH)_2Si_4O_{14}$
$2x - 2 + 4 \times 4 + 14 \times (-2) = 0$
$x = 7$

9. **Ans (4)**

(i) LiCl > NaCl > KCl > RbCl

Anion same so $\boxed{\text{Size of cation} \propto \frac{1}{PP}}$

∴ Polarsing power ∝ covalent character

(ii) Covalency ⟶ it is defined as number of covalent bonds formed by an atom For 'F' covalnency is 1

(iii) $M^{+2} \underset{\text{E.A.of } M^{+3}}{\overset{IP_1 \text{ of } M^{+2}}{\rightleftharpoons}} M^{+3}$

IP_1 of M^{+2} = E.A. of M^{+3}

(iv) $NaCl < MgCl_2 < AlCl_3$
Anion is same

So, lattice energy $\propto \frac{\text{charge}}{\text{size}}$

Charge ↑ LE ↑
Size ↓ LE ↑
Charge is dominant over size

10. **Ans (3)**

Theory Base

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$n\left(\bar{A} \cap \bar{B}\right) = n\left(\overline{A \cup B}\right)$

$= n(U) - n(A \cup B)$

$= 600 - (n(A) + n(B) - n(A \cap B))$

$= 600 - 100 - 200 + 50$

$= 350$

2. **Ans (B)**

$f(x) = ax^2 + 2bx - 5c = 0$ has non real roots then

$ax^2 + 2bx - 5c > 0$ or < 0 for $x \in R$

Given $a + b + c > \frac{9c}{4}$

or $4a + 4b + 4c - 9c > 0$

or $4a + 4b - 5c > 0 \Rightarrow f(2) > 0$, so option B is correct.

3. **Ans (B)**

$x^2 - kx - 7 = 0$

$x^2 - 6x - (k + 1) = 0$

let a common root is a

then $\alpha = 1$ or $k = -6$

∴ $x^2 + 6x - 7 = 0 = (x + 7)(x - 1)$

and $x^2 - 6x + 5 = 0 = (x - 5)(x - 1)$

So uncommon roots are -7 & 5

4. **Ans (C)**

$e^y + y = \ell nx \Rightarrow e^x + x = \ell n(f^{-1}(x))$

$\Rightarrow \ell n(f^{-1}(1)) = e + 1.$

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6. Ans (C)

$$\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{3}{1+(n+2)(n-1)}\right)$$

$$= \sum_{n=1}^{\infty} \tan^{-1}(n+2) - \tan^{-1}(n-1)$$

$\tan^{-1}(n) + \tan^{-1}(n+1) + \tan^{-1}(n+2) - \tan^{-1}(1)$

$- \tan^{-1}(2)$

$\frac{\pi}{2} + \tan^{-1} 3$

7. Ans (A)

$\log_2 \sin x - \log_2 \cos x - \log_2 (1 - \tan^2 x) = -1$

Domain :

$$\left.\begin{array}{c} \sin x > 0 \\ \cos x > 0 \\ 1 - \tan^2 x > 0 \end{array}\right\}$$

$x \in \left(2n\pi, 2n\pi + \frac{\pi}{4}\right)$

$\log_2\left(\frac{\tan x}{1-\tan^2 x}\right) = -1 \Rightarrow \frac{2\tan x}{1-\tan^2 x} = 1$

$\tan 2x = 1$

$2x = n\pi + \frac{\pi}{4} \Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}$

Intersection with Domain : $x = \frac{\pi}{8}$

as $x \in [0, 2\pi)$

8. Ans (A)

$S_n = n(n^2 - 1)$

$\therefore \quad T_n = 3n(n-1)$

$\therefore \quad \frac{1}{T_r} = \frac{1}{3}\left[\frac{1}{r-1} - \frac{1}{r}\right]$

$\therefore \quad \sum_{r=2}^{\infty} \frac{1}{T_r} = \frac{1}{3}$

9. Ans (B)

(A) $\tan 42°(\tan 16°\tan 44°\tan 76°)$

$= \tan 42° \tan 48° = 1$

(B) $(1 - \cot 18°)(1 - \cot 27°)$ is equal to 2

(C) $\prod_{k=1}^{15} \sin\left(\frac{k\pi}{10}\right)$ is equal to 0

$$\sin\frac{\pi}{10}.\sin\frac{2\pi}{10}\ldots\underbrace{\sin\frac{10\pi}{10}}_{0}.\sin\frac{11\pi}{10}\ldots\sin\frac{15\pi}{10}$$

(D) $\cos^2 65° + \cos^2 55° + \cos 65° \cos 55°$

$1 - \sin^2 65° + \cos^2 55° + \frac{2\cos 65° \cos 55°}{2}$

$1 + \cos 120°.\cos 10° + \frac{1}{2}(\cos 120° + \cos 10°)$

$1 - \frac{1}{4} = \frac{3}{4}$

10. Ans (B)

$a = 21$

$d =$ LCM of (4, 5)

$d = 20$

common terms 21, 41, 61, ...

$21 + (n-1) \times 20 \le 417$

$n \le 20.8$

$n = 20$

11. Ans (C)

$$\lim_{x\to\infty} \frac{x^3(1-a) - bx^2 - ax + 1 - b}{x^2 + 1} = 2$$

$1 - a = 0; -b = 2$

12. Ans (D)

If $f(x)$ is continuous at $x = 0$ then

$\lim_{x\to 0} f(x) = f(0)$

$$\Rightarrow \lim_{x\to 0} \frac{\sin 2x - 2\tan x}{\ell n(1+x^3)} = k$$

$$\Rightarrow \lim_{x\to 0} \frac{2\sin x\left(\cos x - \frac{1}{\cos x}\right)}{\ell n(1+x^3)} = k$$

$$\Rightarrow \lim_{x\to 0} \frac{2\sin x\left(-\sin^2 x\right)}{\cos x\, \ell n(1+x^3)} = k$$

$$\Rightarrow \lim_{x\to 0} \frac{-2\sin^3 x}{x^3(\cos x)} \frac{x^3}{\ell n(1+x^3)} = k$$

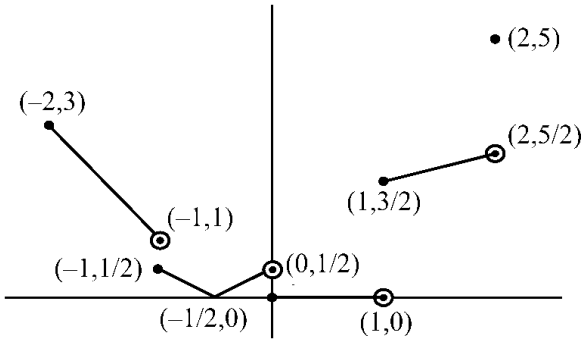
$\Rightarrow -2(1)(1) = k$

$\Rightarrow k = -2$

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13. Ans (C)

Graph of given function



14. Ans (C)

Hence $f(x)$ is continuous at $x = \frac{\pi}{2}$

$$\Rightarrow k = \lim_{x\to\frac{\pi}{2}} \frac{1-\sin x}{(\pi-2x)^2}\cdot\frac{\log\sin x}{\log(1+\pi^2-4\pi x+4x^2)}$$

Put $\alpha = \left(\frac{\pi}{2}-x\right)$

$$k = \lim_{\alpha\to0}\frac{1-\cos\alpha}{(2\alpha)^2}\cdot\frac{\log\cos\alpha}{\log(1+4\alpha^2)}$$

$$= \lim_{\alpha\to0}(1-\cos\alpha)\frac{\log(1+\cos\alpha-1)}{(\cos\alpha-1)}\frac{4\alpha^2}{\log(1+4\alpha^2)}\frac{(\cos\alpha-1)}{(2\alpha)^2 4\alpha^2}$$

$$= \lim_{\alpha\to0}\frac{1-\cos\alpha}{\left(\frac{\alpha^2}{4}\right)}\frac{(\cos\alpha-1)}{\frac{\alpha^2}{4}}\frac{\frac{\alpha^2}{4}\cdot\frac{\alpha^2}{4}}{4\alpha^2.4\alpha^2}$$

$$k = \frac{2.(-2)}{4.4.4.4} = -\frac{1}{64}$$

15. Ans (A)

Let $g^{-1}(x) = f(x)$

$\therefore f(g(x)) = x$

$f'(g(x))\cdot g'(x) = 1$

Put $x = 0$, $f'(g(0))\cdot g'(0) = 1$

$\therefore f'(2)\cdot 6 = 1$

$\therefore \frac{1}{f'(2)} = 6$

16. Ans (C)

As mentioned above, the first term is clearly $\frac{3}{2}$.

For the second term, we first wish to find $\frac{\cos 2x}{\cos 2y} = \frac{2\cos^2 x-1}{2\cos^2 y-1}$. Now we first square the first equation getting $\frac{\sin^2 x}{\sin^2 y} = \frac{1-\cos^2 x}{1-\cos^2 y} = 9$. Squaring the second equation yields $\frac{\cos^2 x}{\cos^2 y} = \frac{1}{4}$. Let $\cos^2 x = a$ and $\cos^2 y = b$. We have the system of equations

$$1 - a = 9 - 9b$$

$$4a = b$$

Multiplying the first equation by 4 yields $4 - 4a = 36$ and so $4 - b = 36 - 36b \Rightarrow b = \frac{32}{35}$. We then find $a = \frac{8}{35}$. Therefore the second fraction ends up being $\frac{\frac{64}{35}-1}{\frac{16}{35}-1} = -\frac{19}{29}$ so that means our desired sum is $\frac{49}{58}$ so the desired sum is 107.

17. Ans (B)

Using LMVT, $f'(x_0) = \frac{f(5)-f(2)}{5-2} = f'(x_0) = \frac{1}{10}$

18. Ans (D)

$$y = \frac{x+6}{x^2-5x+6}$$

$x = 0,\ y = 1$

$$M_T = \frac{dy}{dx} = \frac{(x^2-5x+6)1-(x+6)(2x-5)}{(x^2-5x+6)^2}$$

$$M_T\,]_{(0,1)} = \frac{36}{36} = 1$$

$M_N = -1$

equation $y-1 = -1(x-0)$

$y = -x+1$

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19. Ans (B)

$\phi'(x) = f'(x) - f'(2-x)$

$\because \quad f''(x) = +$

for $x > 2 - x$ i.e. $x > 1$ $f'(x) > f'(2-x)$

for $x < 2 - x$ i.e. $x < 1$ $f'(x) < f'(2-x)$

20. Ans (D)

Using LMVT in $[-2, 5]$

$$-4 \le \frac{f(5) - f(-2)}{7} \le 3$$

$$-28 \le f(5) - f(-2) \le 21$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (26)

$k\, f(k) + 2 = \lambda (x-2)(x-3)(x-4)(x-5)$

$\ldots(1)$

put $x = 0$

we get $\lambda = \frac{1}{60}$

Now put λ in equation (1)

$\Rightarrow kf(k) + 2 \frac{1}{60}(x-2)(x-3)(x-4)(x-5)$

Put $x = 10$

$\Rightarrow 10f(10) + 2 = \frac{1}{60}(8)(7)(6)(5)$

$\Rightarrow 52 - 10f(10) = 52 - 26 = 26$

2. Ans (6)

$f'(x) \ge 0 \ \forall \ x \in R$

$\Rightarrow \ 3x^2 + 4kx + (k^2 + 12) \ge 0 \ \forall \ x \in R$

Put $D \le 0$

$\Rightarrow \ k \in [-6, 6]$

3. Ans (4)

$$\alpha = \frac{n(n+1)}{2}$$

$$\beta = \frac{\sqrt{10}}{3} \frac{n(n+1)(2n+1)}{6},$$

$$\gamma = \left(\frac{n(n+1)}{2}\right)^2$$

$\beta^2 = \alpha\gamma$

$$\Rightarrow \frac{10}{9} \frac{(n(n+1)(2n+1))^2}{36}$$

$$= \frac{n(n+1)}{2} \cdot \left(\frac{n(n+1)}{2}\right)^2$$

$$\frac{10}{9 \times 36} n^2(n+1)^2(2n+1)^2$$

$$= \frac{n(n+1)}{2} \cdot \frac{n^2(n+1)^2}{4}$$

$20(2n+1)^2 = 81n(n+1)$

$20[4n^2 + 4n + 1] = 81n^2 + 81n$

$n^2 + n - 20 = 0$

$(n+5)(n-4) = 0$

$n \ne -5, n = 4, n \in N$

4. Ans (1)

$S = 1 + 2.2 + 3.2^2 + \ldots 100.2^{99}$

$2S = 1.2 + 2.2^2 + \ldots + 100.2^{100}$

$-S = (1 + 2 + 2^2 + \ldots + 2^{99}) - 100.2^{100}$

$S = 1 + 99.(2)^{100}$

$a = 99 \quad ; \quad b = 2 \quad ; \quad c = 100$

$\therefore \frac{c-a}{b} = \frac{1}{2} = 0.50$

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5. Ans (8)

$\alpha + \beta = 6 \qquad \alpha\beta = 12$

$(\alpha - 2)^3 = \alpha^3 - 8 - 3\alpha.2(\alpha - 2)$

$= \alpha(\alpha^2 - 6\alpha + 12) - 8$

$\Rightarrow (\alpha - 2)^3 = -8$

$\Rightarrow \quad (\alpha - 2)^{24} = 8^8$

$\beta^2 - 6\beta + 12 = 0$

$\Rightarrow \quad \beta = \dfrac{-12}{\beta - 6}$

$(\beta - 6)^8 = \dfrac{12^8}{\beta^8}$

$\Rightarrow \dfrac{(\beta - 6)^8}{\alpha^8} = \dfrac{12^8}{(\alpha\beta)^8} = 1$

$\Rightarrow 8^8 - 1 + 1 = 8^8 = a^b$

$\Rightarrow a = 8, b = 8$

$\Rightarrow \dfrac{a + b}{2} = 8$

6. Ans (2)

Let $E = \cos^2\theta - 3\sin2\theta + 3\sin^2\theta + 2$

$= 4 - \cos2\theta - 3\sin2\theta$

$E \in \left[4 - \sqrt{10}, 4 + \sqrt{10}\right]$

The max. value of $\log_{\left(4+\sqrt{10}\right)^2}(E) = \dfrac{1}{2}$

7. Ans (0)

$\lim\limits_{x\to\infty} x\left(\left(1 + \dfrac{3}{x^3}\right)^{1/4} - \left(1 + \dfrac{3}{x^3}\right)^{1/3}\right)$

$\lim\limits_{x\to\infty} x\left(\left(1 + \dfrac{3}{4x^3}\right) - \left(1 + \dfrac{1}{x^3}\right)\right)$

$\lim\limits_{x\to\infty} x\left(-\dfrac{1}{4x^3}\right) = 0$

8. Ans (2)

$\sin^{-1}(\sin3) = \pi - 3$

$\tan^{-1}(\tan3) = 3 - \pi$

so $f(\pi - 3) + f(3 - \pi)$

Let $(\pi - 3) = t$

$f(t) + f(-t) = 2$

9. Ans (9)

f(x) should be ax^3 type function

so $f(x) = (3x - 4)^3 = 27x^3 - 3\cdot9x^2\cdot4 + 3\cdot16\cdot3x - 64$

$\Rightarrow \quad p = -54$

$q = 144$

10. Ans (6)

$\cot^4x - 2(1 + \cot^2x) + (a - 2)^2 + 2 = 0$

$\cot^4x - 2\cot^2x + (a - 2)^2 = 0$

$(\cot^2x - 1)^2 = 1 - (a - 2)^2$

$\geq 0 \qquad \geq 0$

$1 - (a - 2)^2 \geq 0$

$(a - 1)(3 - a) \geq 0$

$1 \leq a \leq 3$

sum $= 1 + 2 + 3 = 6$

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